

Advanced programming Assignment

ROLL NO: CSB24065

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Session: Spring

Year:2026

Question:

Design a library system in Java or Python with:

A base/abstract class LibraryItem (common fields like title, year) An abstract/common method displayInfo()

Create subclasses such as: Book (add author) DVD (add duration/genre)

Your implementation should clearly show:

Use of abstraction (common structure in base class) Method overriding in subclasses Polymorphism using a collection of LibraryItem objects

One additional feature: constructor overloading / default arguments OR class/static counter

Answer:

```
import java.util.ArrayList;
```

```
abstract class LibraryItem {
    protected String title;
    protected int year;
    protected static int count = 0;

    public LibraryItem() {
        this("Unknown", 0);
    }

    public LibraryItem(String title, int year) {
        this.title = title;
        this.year = year;
        count++;
    }
}
```

```
}
```

```
public abstract void displayInfo();
```

```
public static int getCount() {
```

```
    return count;
```

```
}
```

```
}
```

```
class Book extends LibraryItem {
```

```
    private String author;
```

```
    public Book() {
```

```
        this("Unknown", 0, "Unknown");
```

```
}
```

```
    public Book(String title, int year, String author) {
```

```
        super(title, year);
```

```
        this.author = author;
```

```
}
```

```
@Override
```

```
public void displayInfo() {
```

```
    System.out.println("Book Title: " + title);
```

```
    System.out.println("Year: " + year);
```

```
    System.out.println("Author: " + author);
```

```
    System.out.println();
```

```
}
```

```
}
```

```
class DVD extends LibraryItem {
```

```
private int duration;  
private String genre;
```

```
public DVD() {  
    this("Unknown", 0, 0, "Unknown");  
}
```

```
public DVD(String title, int year, int duration, String genre) {  
    super(title, year);  
    this.duration = duration;  
    this.genre = genre;  
}
```

```
@Override
```

```
public void displayInfo() {  
    System.out.println("DVD Title: " + title);  
    System.out.println("Year: " + year);  
    System.out.println("Duration: " + duration + " mins");  
    System.out.println("Genre: " + genre);  
    System.out.println();  
}  
}
```

```
public class LibrarySystem {  
    public static void main(String[] args) {  
        ArrayList<LibraryItem> items = new ArrayList<>();  
  
        items.add(new Book("Java Basics", 2020, "James Gosling"));  
        items.add(new DVD("Inception", 2010, 148, "Sci-Fi"));  
        items.add(new Book("Python Guide", 2022, "Guido van Rossum"));  
    }  
}
```

```

        for (LibraryItem item : items) {
            item.displayInfo();
        }

        System.out.println("Total Library Items: " + LibraryItem.getCount());
    }
}

```

Screenshots of the output:

The screenshot shows an IDE with two tabs: `BankingSystem.java` and `LibrarySystem.java`. The `LibrarySystem.java` tab is active, showing the following code:

```

1 import java.util.ArrayList;
2
3 abstract class LibraryItem {
4     protected String title;
5     protected int year;
6     protected static int count = 0;

```

The terminal window at the bottom shows the execution of the program. The command executed is:

```

PS C:\Users\gouri\OneDrive\Desktop\new_java> cd "c:\Users\gouri\OneDrive\Desktop\new_java\" ; if ($?) { javac LibrarySystem.java } ; if ($?) { java LibrarySystem }

```

The output of the program is:

```

Book Title: Java Basics
Year: 2020
Author: James Gosling

DVD Title: Inception
Year: 2010
Duration: 148 mins
Genre: Sci-fi

Book Title: Python Guide
Year: 2022
Author: Guido van Rossum

Total Library Items: 3
PS C:\Users\gouri\OneDrive\Desktop\new_java>

```